

Q3) (PART I)
How RUP, Prototyping & Agile help manage complexity
& reduce late-stage bug detection cost.

RUP

- ★ Rational Unified process → Iterative & architecture-centric process model w 4 phases.
Inception, Elaboration, Construction & Transition.
- ★ It manages complexity by breaking large projects into multiple iterations, each iteration includes reqs, design, coding & testing
- ★ Elaboration phase focuses on architecture & risk elimination. Since risk & design issues are resolved early, cost of fixing defects later in development is reduced significantly. Continuous testing & incremental integration prevent late stage ~~bug~~ failures.

PROTOTYPING MODEL

This model reduces complexity by first building → quick, working model of system
This helps stakeholders visualize system early, correct misunderstanding & refine reqs.
Thus, the model prevents major req-related defects that typically appear late in Waterfall model.
Misconceptions are caught early → reducing expensive late-stage bug fixing.

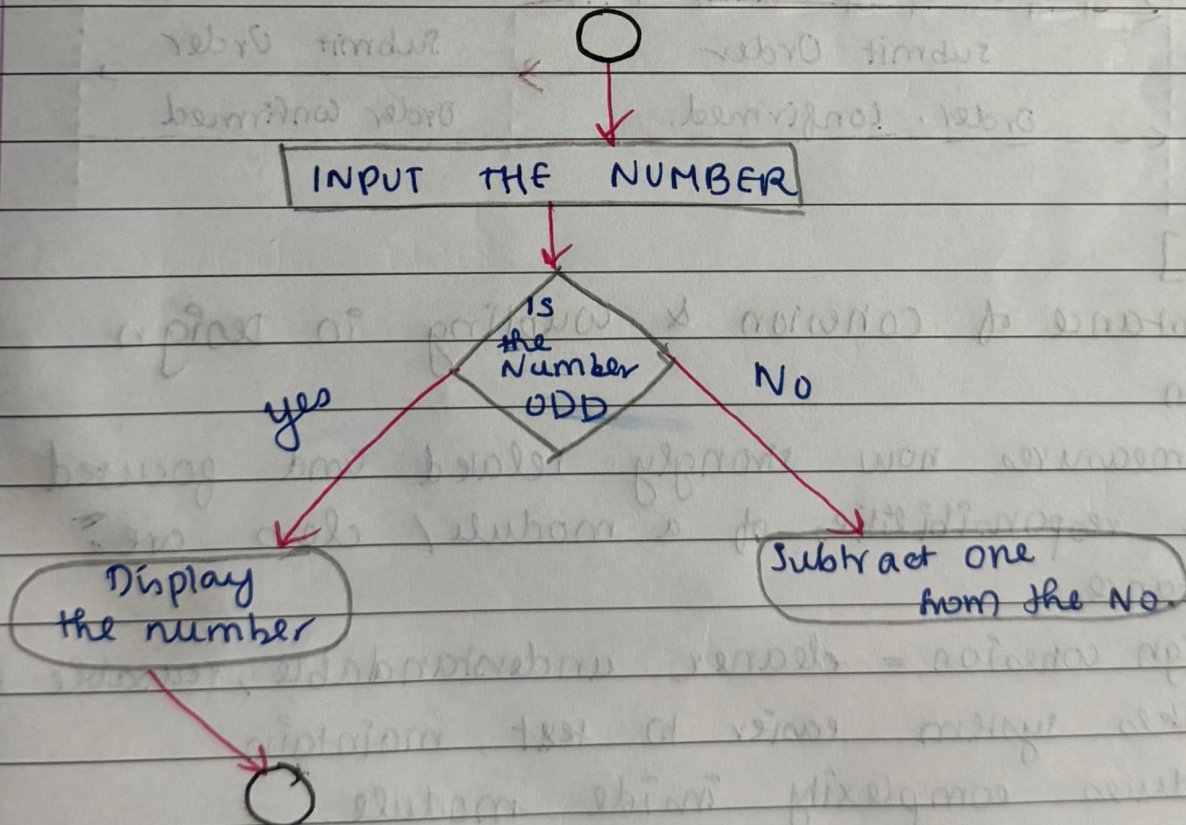
- 5) Deployment (Agile + Waterfall)
Pilot deployment (Agile)
Full-scale deployment after acceptance (Waterfall).
- 6) Maintenance (Agile cycles)
• Continuous updates through mini-sprints.

Q3) [PART II]

UML Activity Diagram

An activity Diagram is an essential part of the unified modelling language that help visualize and model the workflow / business process of system.

Purpose: Show flow of activities from start to end.
help analyst understand overall process by coding.



AGILE MODEL

Agile follows short, time-boxed iteration (sprints) where requirements, design, coding & testing happen together. Complexity is managed by delivering small, incremental features instead of a large monolithic system. Continuous feedback from stakeholder ensures wrong features, unusable designs / reqs changes are caught early. Since testing is integrated into every sprint, bugs are detected immediately, heavily reducing late-stage defect cost.

(PART II)

HYBRID SDLC Model combining Waterfall + Agile.

Govt

Govt. project usually need strict documentation, formal approval, fixed scope & audit trail. [Waterfall strength]
They also benefit from flexibility, faster feedback, & incremental delivery. (Agile strength).

HYBRID MODEL

Requirement Gathering (Waterfall)

- Entire SRS prepared upfront for govt. approval.
- Ensure legal compliance, audit doc^{hm} & clear scope.

High-level Design (Waterfall)

- Architecture & module defined
- Helps handle large system complexity.

Implementation in Iterations [Agile]

- Each module is developed in sprints.
- Daily standups, continuous testing & incremental integration.

Testing [Agile]

- Sprint level testing + Final system testing.
- Ensure early defect detection & high reliability.

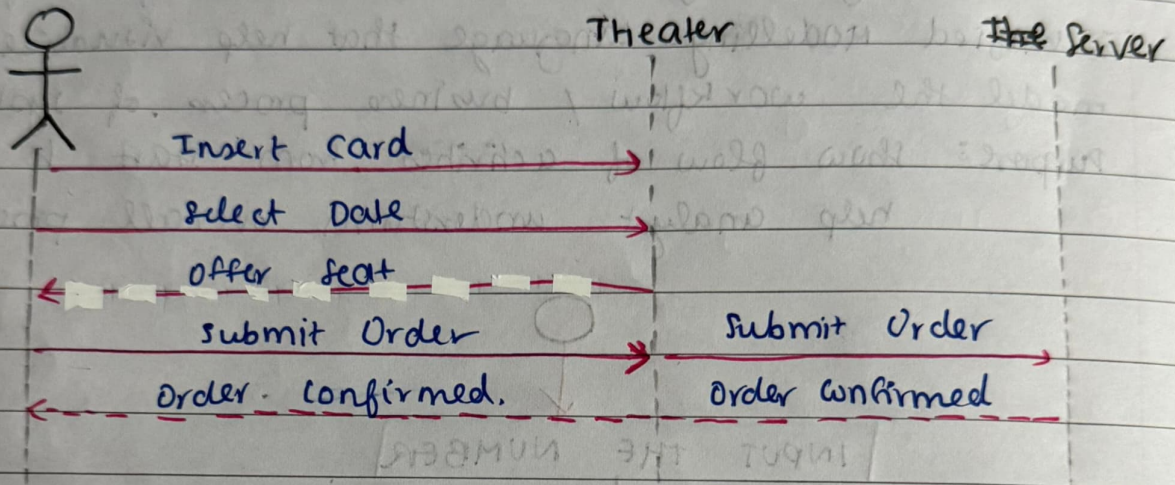
SEQUENCE DIAGRAM

A sequence diagram in UML is used to visualize / model the interaction between object in a time-ordered sequence.

Purpose:

- shows how component/object communicate step by step.
- Help developer understand message flow, request / response & timing.

user interacting w seat reservation system



Q3. [PART 4]

Importance of Cohesion & Coupling in Design

Cohesion

↳ It measures how strongly related and focused the responsibilities of a module / class are.

Importance

- ↳ High cohesion = cleaner, understandable, reusable module.
- ↳ Makes system easier to test, maintain.
- ↳ Reduces complexity inside module.
- ↳ Each module does one job well-defined (Best case)

COUPLING

↳ coupling is an essential design principle that measures how much a module depends on another.

Importance

- ↳ low coupling makes the system more flexible & resilient to change
- ↳ Reduces ripple effect when one module is updated.
- ↳ encourages modularity & maintainability
- ↳ ~~modules depend~~ ensures module depend minimally on each other.